

- 1 (a) Define acceleration.

.....
 [1]

- (b) An Olympic diver stands on a platform above a pool of water, as shown in Fig. 2.1.

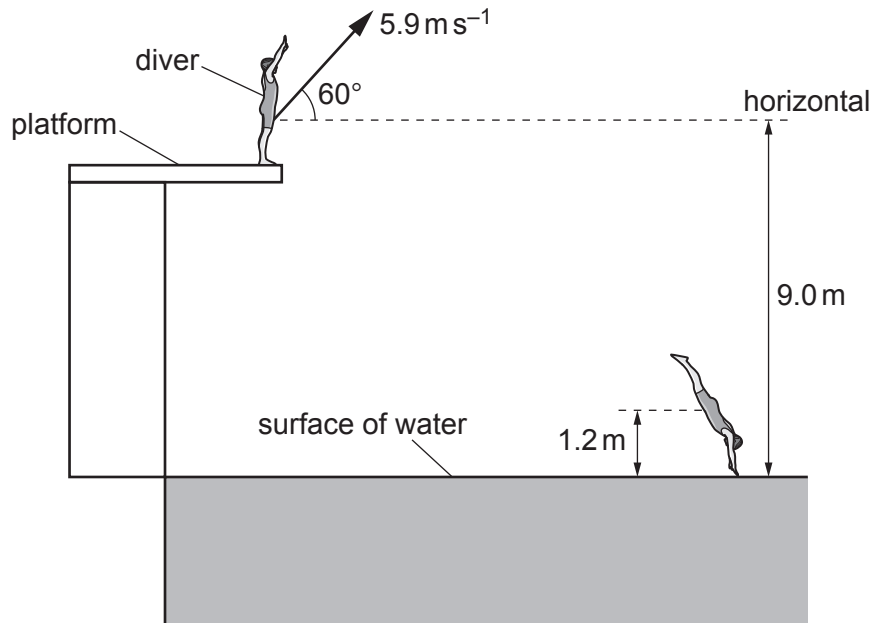


Fig. 2.1 (not to scale)

When the diver is on the platform his centre of gravity is a vertical height of 9.0 m above the surface of the water. The diver jumps from the platform with a velocity of 5.9 m s^{-1} at an angle of 60° to the horizontal.

Air resistance is negligible.

When the diver hits the surface of the water, his centre of gravity is a vertical height of 1.2 m above the surface of the water.

Calculate the speed of the diver at the instant he hits the surface of the water.

speed = m s^{-1} [3]

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(c) The diver in (b) enters the water and decelerates.

(i) Describe and explain the variation of the viscous drag force acting on the diver in the water as he moves downwards.

.....

.....

.....

..... [2]

(ii) The diver has a volume of $7.5 \times 10^{-2} \text{ m}^3$. The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$.

Show that the upthrust acting on the diver when he is entirely underwater is 740 N.

[1]

(iii) At a particular instant when the diver is entirely underwater his horizontal velocity is zero. The viscous drag force acting on him at this instant is 950 N vertically upwards. The diver has mass 78 kg.

Determine the magnitude and direction of the acceleration of the diver.

acceleration = ms^{-2}

direction [4]

[Total: 11]

- 2 (a) Define velocity.

.....
 [1]

- (b) A student throws a ball over a vertical wall of height h , as shown in Fig. 2.1.

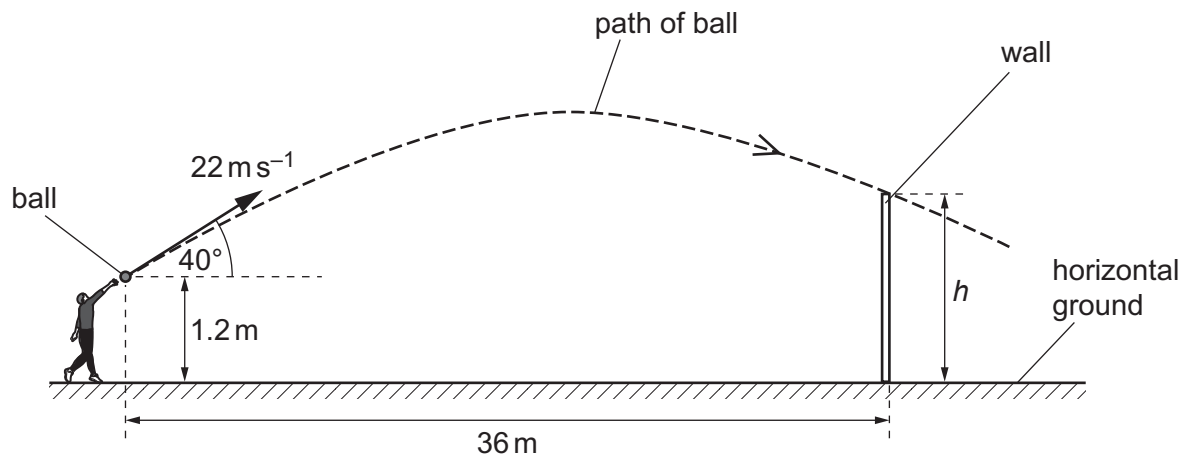


Fig. 2.1 (not to scale)

The ball leaves the hand of the student at a height of 1.2 m above the horizontal ground.
 The ball has an initial velocity of 22 m s^{-1} at an angle of 40° to the horizontal.
 The wall is a horizontal distance of 36 m from where the student releases the ball.

Air resistance is negligible.

- (i) Determine the time taken for the ball to reach the wall.

time taken = s [2]

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- (ii) Calculate the vertical component u of the initial velocity of the ball.

$$u = \dots\dots\dots \text{ms}^{-1} \quad [1]$$

- (iii) The ball just goes over the wall.

Calculate the height h of the wall.

$$h = \dots\dots\dots \text{m} \quad [3]$$

[Total: 7]

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- 3 A skydiver jumps from an aircraft at time $t = 0$ and falls vertically downwards. The variation with t of her velocity v is shown in Fig. 2.1.

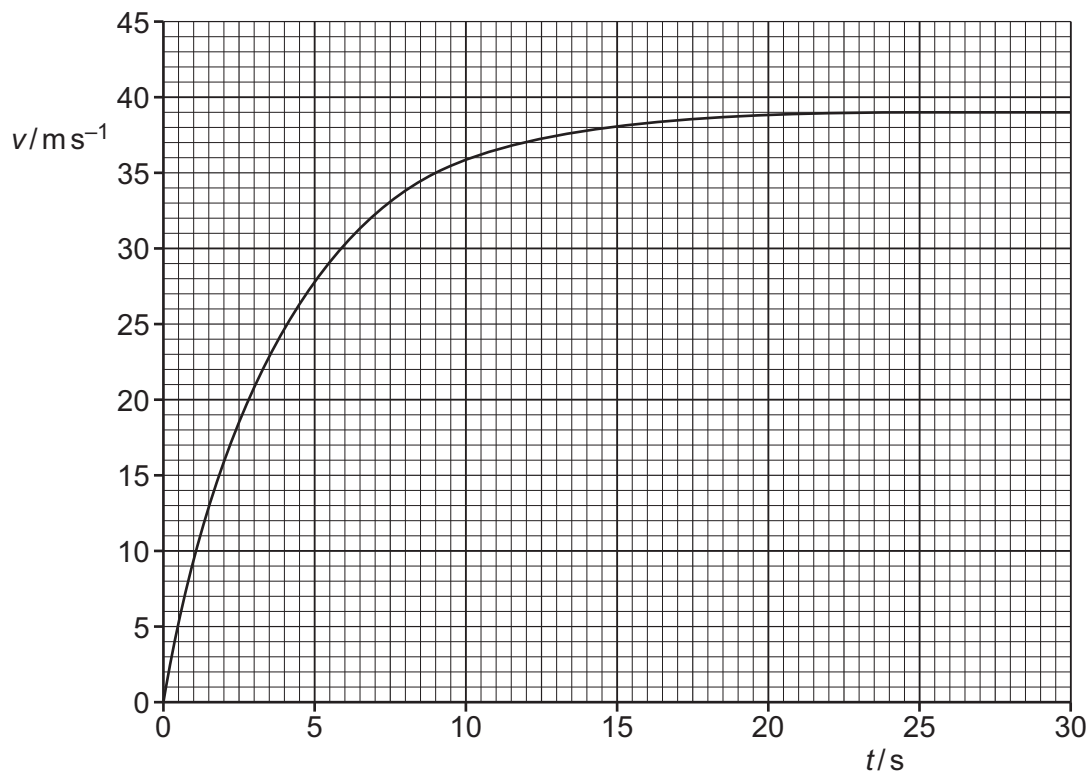


Fig. 2.1

- (a) (i) Using Fig. 2.1, state the terminal velocity of the skydiver.

terminal velocity = ms^{-1} [1]

- (ii) By drawing a suitable line on Fig. 2.1, determine the acceleration of the skydiver at time $t = 9.0\text{ s}$.

acceleration = ms^{-2} [2]

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- (b) The mass of the skydiver and her equipment is 68 kg. The upthrust on the skydiver is negligible.

After reaching terminal velocity, the skydiver opens her parachute at time t_1 . A total drag force of 1800 N acts on the skydiver.

Determine the magnitude and direction of the acceleration of the skydiver at time t_1 .

acceleration =ms⁻²

direction =
[3]

- (c) The parachute is fully open at time t_2 . At a later time t_3 the skydiver reaches a constant velocity of 5.7 ms⁻¹.

- (i) Describe and explain the variation with time of the magnitude of her acceleration between time t_2 and time t_3 .

.....
.....
.....
.....
..... [2]

- (ii) Calculate the change in momentum of the skydiver between time t_1 and time t_3 .

change in momentum = Ns [2]

[Total: 10]

- 4 (a) Define displacement from a point.

.....
 [1]

- (b) An object is projected horizontally at a speed of 6.0 m s^{-1} from a slope, as shown in Fig. 2.1.

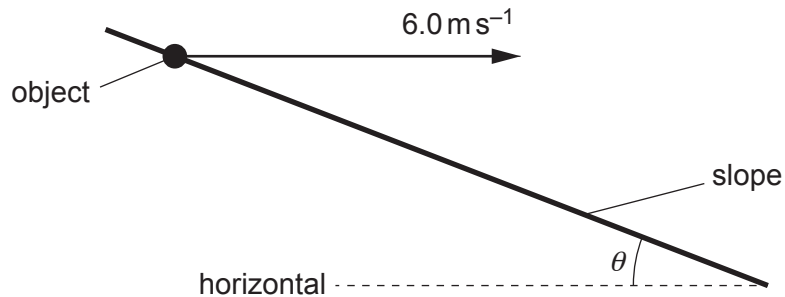


Fig. 2.1 (not to scale)

The slope is at an angle θ to the horizontal. Air resistance is negligible.

The object lands on the slope a time of 0.71 s later and stops without rolling or bouncing.

- (i) Determine the horizontal distance travelled by the object.

distance = m [1]

- (ii) Determine the vertical distance travelled by the object.

distance = m [2]

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(iii) Use your answers in (b)(i) and (b)(ii) to calculate θ .

$$\theta = \dots\dots\dots^\circ \quad [2]$$

(iv) Determine the magnitude of the displacement of the object from its original position.

$$\text{displacement} = \dots\dots\dots \text{ m} \quad [2]$$

(v) By considering energy, calculate the speed of the object just before it lands.

$$\text{speed} = \dots\dots\dots \text{ m s}^{-1} \quad [3]$$

[Total: 11]

- 5 (a) Define acceleration.

.....
 [1]

- (b) A small aircraft is flying horizontally at a speed of 42 m s^{-1} at a height of 63 m above horizontal ground, as shown in Fig. 1.1.

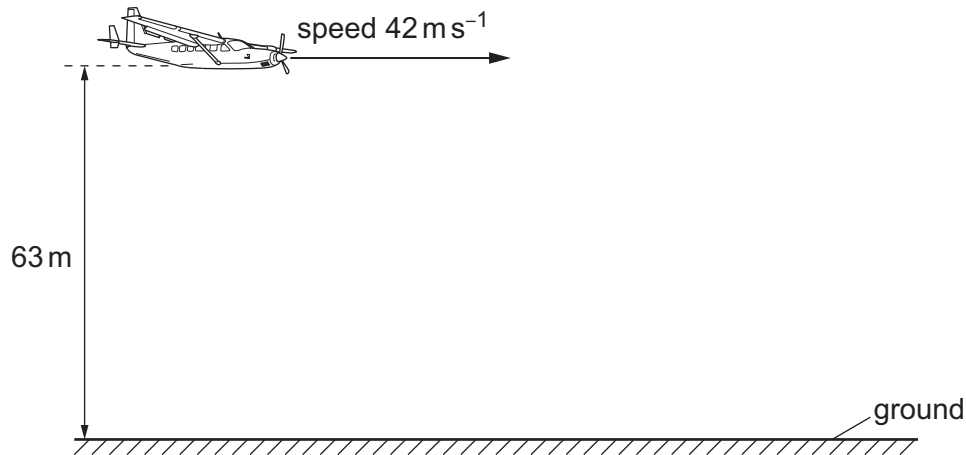


Fig. 1.1

The aircraft drops a small parcel. The parcel is released from the aircraft at the instant shown in Fig. 1.1. Air resistance is negligible.

- (i) On Fig. 1.1, draw a line to show the path of the parcel as it falls from the aircraft to the ground. [1]
- (ii) Calculate the time taken from the instant of release to the instant the parcel reaches the ground.

time = s [2]

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- (iii) Calculate the vertical component of the velocity of the parcel immediately before it reaches the ground.

vertical component of velocity = ms^{-1} [1]

- (iv) Determine the speed at which the parcel reaches the ground.

speed = ms^{-1} [2]

[Total: 7]